

# Iason Kalogiannis

Zografou, Athens, Greece · iasonklgns@gmail.com

## Research Interests

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Theoretical foundations of deep learning, training & learning dynamics, statistical physics of machine learning, and optimization

## Education

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**National Technical University of Athens (NTUA)**

Athens, Greece

*Diploma (M.Eng.) in Electrical and Computer Engineering*

2018 — 2026

- Overall grade: **8.34/10**
- Focus on machine learning, signal processing, and optimization.

## Diploma Thesis

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### Initialization and Optimization of Neural Networks with Ranking Layers

*Supervisor: Prof. Petros Maragos; Research mentor: Dr. Nikos Tsilivis*

- Derived nonsmooth backpropagation rules for morphological, ranking, and linear (MRL) networks using the Clarke subdifferential.
- Used mean-field theory to identify an initialization regime that balances forward and backward signal propagation in deep MRL networks.
- Validated the theoretical predictions on MNIST, showing that the trainable initialization regime narrows toward the region between the two criticalities as network depth increases.

## Selected Coursework

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- Pattern Recognition — 10/10,      Matrix Analysis — 10/10
- Digital Signal Processing — 9/10,      Computer Vision — 9/10
- Calculus of Variations and Optimal Control — 9/10
- Neural Networks and Deep Learning — 10/10

## Experience

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### API Developer — ECE Department Library, NTUA

- Sole developer of a new API for the Electrical and Computer Engineering department library at NTUA.

## Skills

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**Programming:** Python, MATLAB, C/C++

**Languages:** English (Proficient), Greek (Native)